Akash Jose

Computer Science Undergraduate | Software Engineer | AI Enthusiast

akaxh07@gmail.com | +91 7907764791 | Kottayam, Kerala | linkedin.com/in/akash-jose-231954295 | github.com/AkashJos3

leetcode.com/u/Akash_Dev-07

PROFESSIONAL SUMMARY

Computer Science undergraduate with a strong foundation in Java, Python, data structures, and backend development. Built REST APIs and AI-powered applications using FastAPI, PostgreSQL, and Retrieval-Augmented Generation (RAG). Passionate about software engineering and artificial intelligence, with a strong interest in building practical solutions that solve real-world problems.

EDUCATION

Bachelor of Technology (B.Tech) in Computer Science and Engineering 2023 – Present | Kottayam, India
St. Joseph's College of Engineering and Technology (SJCET)
CGPA: 7.2/10
Relevant Coursework: Data Structures & Algorithms, Object-Oriented Programming, Database Management Systems, Operating Systems, Computer Networks.

SKILLS SUMMARY

Programming Languages
Java, Python, C, SQL, JavaScript

Tools
VS Code, Git, GitHub, Postman, Docker, Maven

Backend
Spring Boot, Spring Security, FastAPI, Flask, REST APIs, JWT Authentication

Computer Science Fundamentals
Data Structures & Algorithms, Object-Oriented Programming, DBMS, Operating Systems, Computer Networks

Databases
PostgreSQL, MySQL, Supabase

Artificial Intelligence
Retrieval-Augmented Generation (RAG), Prompt Engineering, Vector Databases (ChromaDB), Gemini API

PROJECTS

Expense Tracker Backend API 06/2026 – 08/2026
Tech Stack: Java, Spring Boot, Spring Security, PostgreSQL, Spring Data JPA, JWT
- Built a secure **Spring Boot REST API** for expense management with CRUD operations, pagination, sorting, dynamic filtering, and PostgreSQL persistence.
- Implemented **JWT authentication and user-specific authorization** using Spring Security and BCrypt, with DTO-based architecture and global exception handling.
- Added **JUnit/Mockito/MockMvc testing, Docker configuration, and GitHub Actions CI** with automated PostgreSQL-backed builds and tests.

AI Study Assistant using Retrieval-Augmented Generation (RAG) 01/2026 – 07/2026
Tech Stack: Python, FastAPI, PostgreSQL, ChromaDB, Gemini API
- Developed an AI-powered study assistant enabling students to upload PDFs and interact with study materials through natural-language queries.
- Implemented a Retrieval-Augmented Generation (RAG) pipeline using **ChromaDB and Gemini API** for context-aware document question answering.
- Integrated **AI-generated summaries, quizzes, and flashcards** to provide a personalized and interactive learning experience.

Afterbell – Student Part-Time Job Portal 08/2025 – 03/2026
Tech Stack: Python, FastAPI, PostgreSQL, Supabase
- Developed a backend platform that helps students discover and apply for local part-time job opportunities.
- Built REST APIs for user authentication, job posting, job applications, and employer management.
- Implemented role-based access for students, employers, and administrators using Supabase Authentication.

ACHIEVEMENTS & ACTIVITIES

Asthra Tech Fest – SJCET 2023 – 2025
Served as a coordinator, question setter, and event organizer for Root Access and Byte Arena coding competitions.

First Prize – Tekron Cipher Chase 2025
Secured first place in a college-level cryptography and logic-based problem-solving competition by solving technical challenges under time constraints.

NASA Space Apps Challenge (Participant) 2024
Collaborated with a team during the NASA Space Apps Challenge to design and develop a technology-based solution for a real-world space and Earth science problem.

CERTIFICATIONS

Python for Data Science , — *NPTEL (IIT Madras)*

Java Programming Fundamentals — Infosys Springboard

Data Structures and Algorithms — Infosys Springboard

Software Engineering — Infosys Springboard